ICT2142 · Database Systems

Core Concepts: Normalization and Integrity

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

Normalization reduces update anomalies by storing each fact in one appropriate place. First normal form avoids repeating groups; higher forms separate facts that depend on different keys. Constraints express data rules at the database boundary.

2. Example and reasoning

If department name is repeated in every course row, a department rename requires many updates. A department table with course foreign keys stores that fact once and makes referential integrity enforceable.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “core concepts: normalization and integrity” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.